

Muhammad Bilal Qamar

Software Engineer | CS Student

+92 329 3313330 | [Email](#) | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

PROFILE

Motivated 4th-semester Computer Science student at The Superior University with a passion for building practical software and exploring emerging technologies. Experienced in full-stack web development, systems documentation, and AI/ML model training. Currently serving as Softec Ambassador for my campus, and actively seeking an internship where I can apply my technical skills and enthusiasm for problem-solving.

EXPERIENCE

Software Engineering Intern — Glosix Systems

June 2026 – Present

- Developing full-stack web applications, building responsive user interfaces with React and Tailwind CSS on the frontend.
- Building and maintaining backend services and REST APIs using .NET / ASP.NET Core Web API.
- Designing and querying relational databases with SQL Server and MySQL, using LINQ and Entity Framework Core for data access.

SKILLS

Frontend: JavaScript, React, Tailwind CSS, Bootstrap, HTML, CSS, jQuery

Backend & API: .NET / ASP.NET Core Web API (ASP.NET 9), C#, C++, Python, PHP, SignalR, JWT Authentication

Database: SQL Server, MySQL, LINQ, Entity Framework Core

AI / ML: LangGraph, LangChain, Retrieval-Augmented Generation (RAG), Groq LLM API, HuggingFace, Multi-Agent Systems

Cloud & DevOps: Cloudflare R2 (Object Storage), AWS S3, Vercel, Render, GitHub Actions, Git / GitHub

PROJECTS

ShopCo — E-Commerce Store

- Built a full-stack e-commerce platform with a React 19 + Tailwind CSS storefront and a .NET / ASP.NET Core Web API backend, covering product browsing, cart, checkout, and order flows.
- Implemented JWT authentication and role-based authorization (Admin/Customer), with an admin dashboard for product management and Cloudflare R2 for image storage.
- Built real-time customer support chat with SignalR (WebSockets) plus an AI shopping assistant powered by the Groq LLM API, scoped per-user via JWT claims.

RAG PDF Analyzer — Chat with your PDFs

- Built a full-stack RAG (Retrieval-Augmented Generation) platform with FastAPI, React, and Supabase that lets users upload PDFs and query them through an AI chat interface with source-page citations.
- Engineered a document-processing pipeline using LangChain text splitters and batched HuggingFace embeddings, storing vector data in Supabase for fast semantic retrieval.
- Integrated Groq LLM inference with JWT-based user authentication, and designed memory-safe batch processing to prevent out-of-memory failures on free-tier deployments.

Omnimedia — Writes your blog

- Built an AI-powered blog content generation pipeline with LangGraph, FastAPI, and React, using a multi-agent architecture (researcher, SEO, writer, and editor agents).
- Designed parallel agent execution (fan-out/fan-in) for research and SEO analysis feeding into a writer agent, plus a human-in-the-loop approval and revision workflow.
- Implemented MemGPT-style tiered memory to keep long revision chains bounded, with per-request Groq API keys handled securely without server-side persistence.

voyagecraft — Plan your trip

- Built a multi-agent trip-planning app using LangGraph orchestration and Groq's Llama 3.3 70B model, coordinating discovery, itinerary, and budget agents through a FastAPI backend and React frontend.
- Designed a real-time disruption-simulation engine that dynamically re-plans itineraries in response to live events (e.g., weather, delays), with per-session state management.
- Configured multi-environment CORS handling (Vercel preview + production) using origin-regex matching for seamless frontend-backend integration and deployment.

Fintrack — Personal Finance Web Application

Full-Stack Web Development – Under Development

C++ OOP Projects

- Bank Management System — multi-account banking system with OOP design ([GitHub](#)).
- Vehicle Management System — fleet tracking application a simple project([GitHub](#)).
- Taxation System — rule-based tax computation engine built in C++ ([GitHub](#)).

EDUCATION

The Superior University, Lahore

Bachelor of Science in Computer Science (BSCS)

August 2024 – Present

Superior College, Lahore

Intermediate — ICS (Intermediate in Computer Science)

2022 – 2024

Unique Group of Institutions

Matriculation (Secondary School Certificate)

2020 – 2022

EXTRACURRICULAR

Softec Ambassador

The Superior University Campus

2024 – Present

- Representing the campus at Softec — one of Pakistan's largest student-run computing competitions — promoting technical events and student engagement.